

3(a)(i)	<i>if only symbols are used then both have to be clear and correct</i> alpha / α, globin / (polypeptide) chain <u>and</u> beta / β, globin / (polypeptide) chain ; I alpha and beta polypeptides unqualified	1
3(a)(ii)	haem ; A heme / prosthetic group I porphyrin ring / iron / Fe / Fe²⁺	1
3(a)(iii)	primary (structure) / sequence of amino acids (in a polypeptide) ;	1

Question	Answer	Marks
3(b)	any three from: haemoglobin, combines with / binds to / reacts with, carbon dioxide ; carbon dioxide reacts with (terminal), amine group(s) / -NH_2 / -NH, of (each), chain / globin / polypeptide ; R R-group A forms a bond with to form carbaminohaemoglobin ; carbon dioxide is bound / AW, to Hb until in, region of low $p\text{CO}_2$ or high $p\text{O}_2$ / pulmonary circulation / lungs / alveoli ; A (carbaminohaemoglobin) dissociates / unbinds, in the lungs / etc.	3
3(c)	alveolar capillaries / capillaries around alveoli (in the lungs) ; I 'capillaries in the lungs' / 'pulmonary capillaries'	1
3(d)(i)	any two from: curve shifts to the right (and curve is lower) ; haemoglobin is less saturated (at every $p\text{O}_2$) / higher $p\text{O}_2$ required to reach same % saturation as low [DPG] ; A data P_{50} is higher / increases ; A described, e.g. $p\text{O}_2$ to give 50% saturated A use of data for P_{50} – 3.2 kPa to 4.6 kPa (units used at least once) I Bohr effect	2
3(d)(ii)	less oxygen is available (for tissues) ; haemoglobin has a high(er) affinity for oxygen ;	2